Recall: 隐函数之偏导数:

方程组情形: $\begin{cases} F_1(x,y,z)=0 \\ F_2(x,y,z)=0 \end{cases}$ F_1, F_2: 可微的

若 $\forall\ x\in I$, $\exists \begin{cases} y=y(x) \\ z=z(x) \end{cases}$ $x\in I$, 且 $y(x)$, $z(x)$ 可微, 则

$$\Longrightarrow \begin{cases} F_1(x, y(x), z(x)) \equiv 0, \\ F_2(x, y(x), z(x)) \equiv 0, \end{cases} \quad x\in I.$$

两边对 x 求导:

$$\begin{cases} F_{1x}+F_{1y}\cdot y'(x)+F_{1z}\cdot z'(x)=0, \\ F_{2x}+F_{2y}\cdot y'(x)+F_{2z}\cdot z'(x)=0. \end{cases}$$

$$\Longrightarrow \begin{cases} F_{1y}\cdot y'(x)+F_{1z}\cdot z'(x)=-F_{1x}, \\ F_{2y}\cdot y'(x)+F_{2z}\cdot z'(x)=-F_{2x}. \end{cases}$$

若 $J=\dfrac{\partial(F_1,F_2)}{\partial(y,z)} \triangleq \begin{vmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{vmatrix} \neq 0$, 则由 Cramer 法则:

$$y'(x)=\frac{1}{J}\begin{vmatrix} -F_{1x} & F_{1z} \\ -F_{2x} & F_{2z} \end{vmatrix}=\frac{1}{J}\begin{vmatrix} F_{1z} & F_{1x} \\ F_{2z} & F_{2x} \end{vmatrix}=\frac{1}{J}\,\frac{\partial(F_1,F_2)}{\partial(z,x)}.$$

其中 $\dfrac{\partial(F_1,F_2)}{\partial(z,x)} \triangleq \begin{vmatrix} F_{1z} & F_{1x} \\ F_{2z} & F_{2x} \end{vmatrix}$

同理: $z'=\dfrac{1}{J}\,\dfrac{\partial(F_1,F_2)}{\partial(x,y)}$.

特点： 分母上： $J=\dfrac{\partial(F_1,F_2)}{\partial(y,z)}\neq 0$;

$y'(x)$ 之分子上： $\dfrac{\partial(F_1,F_2)}{\partial(z,x)}$ —— 没有 y;

$z'(x)$ 之分子上： $\dfrac{\partial(F_1,F_2)}{\partial(x,y)}$ —— 没有 z.

轮换： $x \quad y \quad z \quad x \quad y \quad z, \quad \cdots$

例 设 $\begin{cases} x+y+z+u+v=1, \\ x^2+y^2+z^2+u^2+v^2=2, \end{cases}$ 求 $u_x,\ v_x,\ u_{xx}\,(=u_{x^2}),\ v_{xx}\,(=v_{x^2})$.

解：本题是把 u, v 看成 (x,y,z) 之函数，即

$u=u(x,y,z),\quad v=v(x,y,z)$

方程组两边对 x 求偏导数：

$$\begin{cases} 1+u_x+v_x=0 \\ \cancel{2}x+\cancel{2}uu_x+\cancel{2}vv_x=0 \end{cases} \qquad J=\begin{vmatrix} 1 & 1 \\ u & v \end{vmatrix}=v-u\neq 0$$

$$\therefore\ u_x=\frac{1}{J}\begin{vmatrix} -1 & 1 \\ -x & v \end{vmatrix}=\frac{x-v}{v-u};$$

$$v_x=\frac{1}{J}\begin{vmatrix} 1 & -1 \\ u & -x \end{vmatrix}=\frac{u-x}{v-u}$$

$$\begin{cases} u_x+v_x=-1 \\ uu_x+vv_x=-x \end{cases}$$

两边再对 x 求偏导数：

$$\begin{cases} u_{xx}+v_{xx}=0; \\ uu_{xx}+vv_{xx}+u_x^2+v_x^2=-1 \end{cases}$$

$$u_{xx}=\frac{1}{J}\begin{vmatrix}0 & 1\\ -1-u_x^2-v_x^2 & v\end{vmatrix}=\frac{1}{J}(1+u_x^2+v_x^2).$$

$$=\frac{1+\left(\frac{x-v}{v-u}\right)^2+\left(\frac{x-v}{v-u}\right)^2}{v-u}=\frac{(v-u)^2+(x-v)^2+(x-u)^2}{(v-u)^3}.$$

同理

$$v_{xx}=\frac{1}{J}\begin{vmatrix}1 & 0\\ u & -1-u_x^2-v_x^2\end{vmatrix}=\frac{1}{J}(-1-u_x^2-v_x^2)$$

$$=\frac{(v-u)^2+(x-v)^2+(x-u)^2}{(u-v)^3}.\quad \#$$

例 设 $u=x+y$, $v=x-y$, $w=xy-z(x,y)$, 把方程

$$\frac{\partial^2 z}{\partial x^2}+2\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial^2 z}{\partial y^2}=0$$

$z_{xy}=z_{yx}$

化为 $w=w(u,v)$ 之方程.

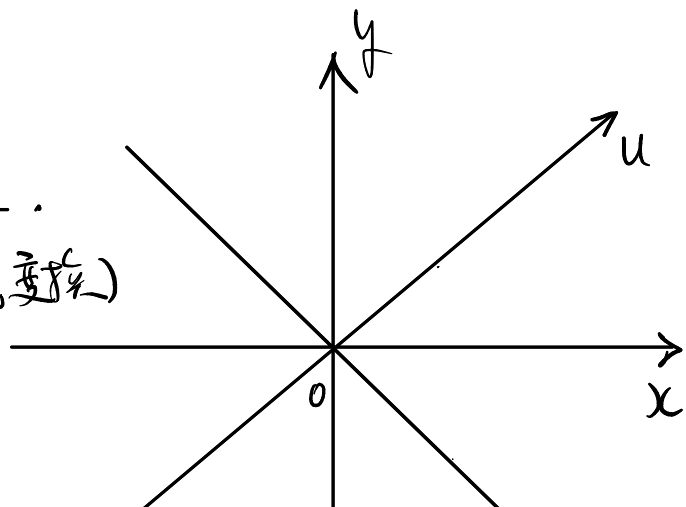
(含有未知函数偏导数之方程: 偏微分方程)

解: $z=xy-w(x+y,\ x-y)$

$=z(x(u,v),\ y(u,v))$

$$\begin{cases}u+v=2x \Rightarrow \\ u-v=2y \Rightarrow\end{cases}\begin{cases}x=\frac{u+v}{2}\\ y=\frac{u-v}{2}.\end{cases}$$

(旋转伸缩变换)



$z=xy-w$:

$$z_x=y-w_u\cdot u_x-w_v\cdot v_x$$

$$=y-w_u\cdot 1-w_v\cdot 1$$

$$=y-w_u-w_v$$

$$z_y = x - W_u \cdot u_y - W_v \cdot v_y = x - W_u \cdot 1 - W_v (-1)$$

$$= x - W_u + W_v .$$

$$z_{xx} = (y - W_u - W_v)_x = 0 - W_{uu} \cdot u_x - W_{uv} \cdot v_x$$

$$- W_{vu} \cdot u_x - W_{vv} \cdot v_x$$

$$= - W_{uu} - 2 W_{uv} - W_{vv}$$

$$z_{xy} = (y - W_u - W_v)_y = 1 - W_{uu} \cdot u_y - W_{uv} \cdot \underbrace{v_y}_{=-1}$$

$$- W_{vu} \cdot u_y - W_{vv} \cdot v_y$$

$$= 1 - W_{uu} + \cancel{W_{uv} - W_{vu}} + W_{vv}$$

$$= 1 - W_{uu} + W_{vv} .$$

$$z_{yy} = (x - W_u + W_v)_y = 0 - W_{uu} \cdot \underbrace{u_y}_{=1} - W_{uv} \cdot \underbrace{v_y}_{=-1}$$

$$+ W_{vu} \cdot u_y + W_{vv} \cdot v_y$$

$$= - W_{uu} + W_{uv} + W_{uv} - W_{vv}$$

$$= - W_{uu} + 2 W_{uv} - W_{vv}$$

$$0 = \frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$$

$$= - W_{uu} - \cancel{2 W_{uv}} - \cancel{W_{vv}}$$

$$+ 2 (1 - W_{uu} + \cancel{W_{vv}})$$

$$+ (- W_{uu} + \cancel{2 W_{uv}} - \cancel{W_{vv}})$$

$$= - 4 W_{uu} + 2 = 0$$

$\therefore \quad 2W_{uu} = 1.$ #

$$\frac{\partial^2 W}{\partial u^2} = \frac{1}{2} :$$

$$\Downarrow$$

$$(W_u)_u = \frac{1}{2} \implies W_u = \frac{1}{2}u + \varphi(v) :$$

$$\therefore W = \frac{1}{4}u^2 + \varphi(v)\,u + \psi(v)$$

$\varphi(v)$: 任意二阶连续可导
$\psi(v)$

$$z_{xx} + 2z_{xy} + z_{yy} = 0$$

解 $z(x, y)$: ~~任意常数~~ ?

$$\implies z = xy - W :$$

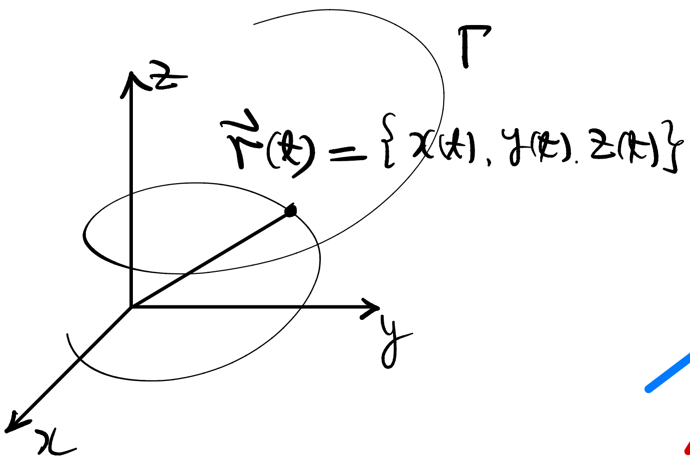
第十五章 偏导数之应用

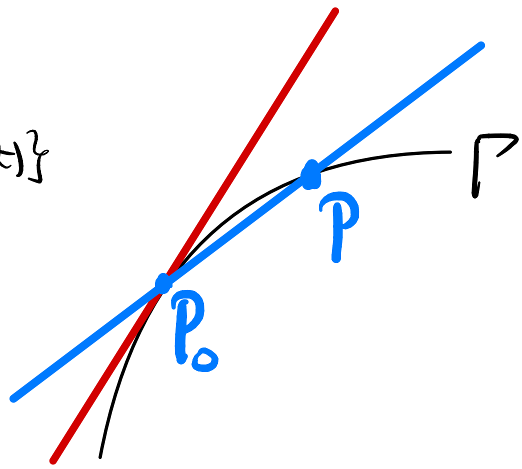
§15.1 空间曲线之切线和法平面

空间一质点随时间运动，其位置坐标是时间之函数。其运动轨迹（坐标）可表示为 $\vec{r}(t)$：

$$\vec{r}(t):(\alpha,\beta)\longrightarrow\mathbb{R}^3$$
$$t\longmapsto(x(t),y(t),z(t))$$

轨迹是一条曲线 Γ：$\vec{r}(t)=\{x(t),y(t),z(t)\}$.





称空间光滑曲线 Γ：$\vec{r}(t)=\{x=x(t),y=y(t),z=z(t)\}$，$t\in(\alpha,\beta)$ 是"光滑"的，是指 Γ 上每点都有"切线"，且切线在曲线上"连续变化"

切线之定义：Γ 上定点 $P_0(x_0,y_0,z_0)=(x(t_0),y(t_0),z(t_0))$，$t_0\in(\alpha,\beta)$，$\Gamma$ 上另一点动点 $P(x,y,z)=(x(t),y(t),z(t))$.

过 P_0,P 之直线称为割线，其方程为：

$$\frac{x-x_0}{x(t)-x_0(t_0)}=\frac{y-y_0}{y(t)-y_0(t_0)}=\frac{z-z_0}{z(t)-z_0(t_0)}$$

P在曲线Γ上连续变化到P_0(即$t\to t_0$)，割线趋于一个极限位置，称为曲线Γ过P_0点之切线.

割线方程两边分母同除以$t-t_0$:

$$\frac{x-x_0}{\frac{x(t)-x_0(t_0)}{t-t_0}}=\frac{y-y_0}{\frac{y(t)-y_0(t_0)}{t-t_0}}=\frac{z-z_0}{\frac{z(t)-z_0(t_0)}{t-t_0}}$$

如果$x(t)$, $y(t)$, $z(t)$在t_0可导且$x'(t_0)$, $y'(t_0)$, $z'(t_0)$不全为零，则令$t\to t_0$，得割线P_0P之极限位置，即切线的方程为:

$$\frac{x-x_0}{x'(t_0)}=\frac{y-y_0}{y'(t_0)}=\frac{z-z_0}{z'(t_0)}$$

$\vec{r}\,'(t_0)\triangleq\{x'(t_0), y'(t_0), z'(t_0)\}\neq\vec{0}$ 是切线之方向.

其单位向量 $\vec{r}_0\,'(t_0)=\dfrac{\vec{r}\,'(t_0)}{|\vec{r}\,'(t_0)|}$ 称为(单位)切向量

曲线Γ: $\vec{r}(t)=\{x(t), y(t), z(t)\}$ 是一光滑曲线

$\overset{\text{定义}}{\Longleftrightarrow}$ $\forall t$, Γ有切线，且切线关于t连续变化

$\Updownarrow$ $\Updownarrow$

$\Longleftrightarrow$ $\vec{r}\,'(t)=\{x'(t), y'(t), z'(t)\}\neq\vec{0}$ 且 $\vec{r}\,'(t)$连续,

即 曲线Γ光滑 $\Leftrightarrow x(t), y(t), z(t) \in C^1(\alpha,\beta)$ 且

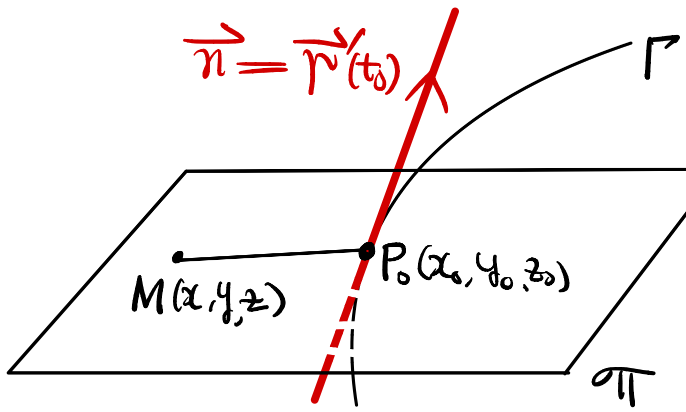
$$\vec{r}\,'(t) = \{x'(t), y'(t), z'(t)\} \neq \vec{0}.$$

$\in C^1(\alpha,\beta)$ 且 $\neq 0$

$t_0\in(\alpha,\beta)$：在 t_0 处，即 $P_0(x_0=x(t_0),\ y_0=y(t_0),\ z_0=z(t_0))\in\Gamma$ 处

曲线的切线方程为

$$\frac{x-x_0}{x'(t_0)} = \frac{y-y_0}{y'(t_0)} = \frac{z-z_0}{z'(t_0)}.$$



过切点P_0且与切线垂直的平面称为法平面.

$\vec{n}=\vec{r}\,'(t_0)$：法方向.

$\forall M(x,y,z)\in\pi \Longleftrightarrow \overrightarrow{P_0M}\perp\vec{n}$ 即 $\vec{n}\cdot\overrightarrow{P_0M}=0$

$$\Longleftrightarrow \{x'(t_0), y'(t_0), z'(t_0)\}\cdot\{x-x_0, y-y_0, z-z_0\}=0$$

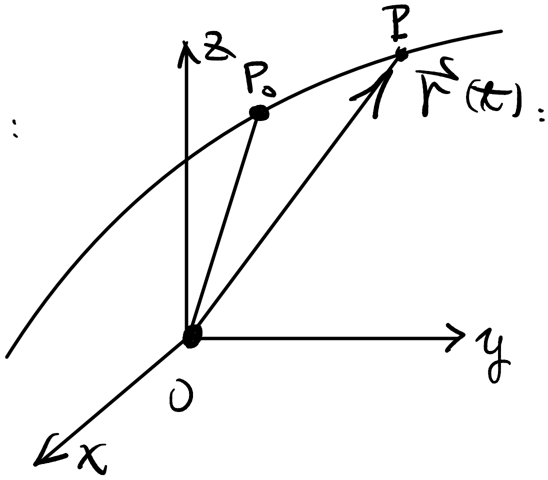
即

$$x'(t_0)(x-x_0)+y'(t_0)(y-y_0)+z'(t_0)\cdot(z-z_0)=0$$

———— 法平面方程.

质点在空间中的运动轨迹：

$\vec{r}(t)=\{x(t), y(t), z(t)\}$;



时间从 $t_0 \to t=t_0+\Delta t$;

质点位置从 $P_0(x_0, y_0, z_0)$

$=(x(t_0), y(t_0), z(t_0))$ 运动

到 $P(x(t), y(t), z(t))$，发生位移

$$\vec{\Delta r}=\overrightarrow{P_0P}=\vec{r}(t)-\vec{r}(t_0)$$

$$=\{x(t)-x(t_0),\ y(t)-y(t_0),\ z(t)-z(t_0)\}$$

平均速度为：

$$\frac{\overrightarrow{P_0P}}{\Delta t}=\left\{\frac{x(t)-x(t_0)}{\Delta t}, \frac{y(t)-y(t_0)}{\Delta t}, \frac{z(t)-z(t_0)}{\Delta t}\right\}$$

令 $\Delta t\to 0$ 得

$$\vec{v}(t)=\lim_{\Delta t\to 0}\frac{\overrightarrow{P_0P}}{\Delta t}$$

$$=\lim_{\Delta t\to 0}\left\{\frac{x(t)-x(t_0)}{\Delta t}, \frac{y(t)-y(t_0)}{\Delta t}, \frac{z(t)-z(t_0)}{\Delta t}\right\}$$

$$=\{x'(t_0), y'(t_0), z'(t_0)\}.$$

这是质点在 t_0 处的瞬时速度.

$\forall t$: 切线方向即运动的方向.

$\vec{v}(t)=\vec{r}'(t)=\{x'(t),\ y'(t),\ z'(t)\}.\ (\neq\vec{0})$

若空间曲线Γ之参数方程为：

$$\begin{cases}x=x\\ y=y(x)\\ z=z(x)\end{cases}\quad x\in(a,b).\ y(x),\ z(x)\in C^1(a,b).$$

$\Gamma:\ \vec{r}(x)=\{x,\ y(x),\ z(x)\}.\qquad x\in(a,b).$

切线方向：$\vec{r}'(x)=\{1,\ y'(x),\ z'(x)\},\ \neq\vec{0}.$

在 $x_0\in(a,b)$ 的切线方程：

$$\frac{x-x_0}{1}=\frac{y-y_0}{y'(x_0)}=\frac{z-z_0}{z'(x_0)},\qquad 其中\ \begin{matrix}y(x_0)=y_0\\ z(x_0)=z_0\end{matrix}$$

法平面：

$$1\cdot(x-x_0)+y'(x_0)(y-y_0)+z'(x_0)(z-z_0)=0$$

若曲线是两个曲面之交线时：

$$\Gamma:\ \begin{cases}F_1(x,y,z)=0,\\ F_2(x,y,z)=0,\end{cases}\qquad F_1,\ F_2\ 连续可微，$$

如果该方程组决定了隐函数组 $\begin{cases}y=y(x),\\ z=z(x),\end{cases}\ x\in I.$

且 $y(x),\ z(x)$ 连续可微，则 Γ 之参数方程为

$$\begin{cases}x=x\\ y=y(x)\\ z=z(x)\end{cases}$$

$$\vec{r}(x)=\{x,\ \underline{y(x)},\ \underline{z(x)}\}.$$

$$J=\frac{\partial(F_1,F_2)}{\partial(y,z)}\neq 0 \Longrightarrow$$

$$y'(x)=\frac{\dfrac{\partial(F_1,F_2)}{\partial(z,x)}}{J}$$

$$z'(x)=\frac{\dfrac{\partial(F_1,F_2)}{\partial(x,y)}}{J}$$

$$\vec{r}'(x)=\{1,\ y'(x),\ z'(x)\}$$

$$=\left\{1,\ \frac{1}{J}\frac{\partial(F_1,F_2)}{\partial(z,x)},\ \frac{1}{J}\frac{\partial(F_1,F_2)}{\partial(x,y)}\right\}$$

$$=\frac{1}{J}\left\{\frac{\partial(F_1,F_2)}{\partial(y,z)},\ \frac{\partial(F_1,F_2)}{\partial(z,x)},\ \frac{\partial(F_1,F_2)}{\partial(x,y)}\right\}$$

切线方向：

$$J\vec{r}'(x)=\left\{\frac{\partial(F_1,F_2)}{\underbrace{\partial(y,z)}},\ \frac{\partial(F_1,F_2)}{\underbrace{\partial(z,x)}},\ \frac{\partial(F_1,F_2)}{\underbrace{\partial(x,y)}}\right\}.$$

依次缺了 x, y, z;

顺序： x, y, z, x, y, z